Evidence – Adaptive Path Planning for Autonomous Vehicles

E01 – Customer Evidence
“The vehicle needs to quickly adapt its path when obstacles or road conditions change unexpectedly.”

E02 – Real Occurrence
“Sudden cattle crossing on an unstructured Indian road requiring the autonomous vehicle to detect the obstacle and replan a safe path in real time.”

E03 – Existing System / Workflow
“Existing autonomous-navigation systems can use sensor data, maps, obstacle detection, and route-planning algorithms to generate paths in known or changing environments.”

E04 – Testing Scenarios
“Sudden pedestrian crossing, vehicle cut-in, lane blocked by construction, dense traffic, temporary road closure, sensor noise / partial occlusion.”

E05 – Validation Metrics
“Collision / near-miss rate, re-planning latency, path length, travel time, prediction accuracy, CPU / memory usage.”

E06 – Key Risks
“Sensor delay, false detections, rare edge cases, over-aggressive re-planning, computation limits, sim-to-real gap.”

E07 – Safety Constraint
“No real-road testing or personal-data collection will occur without approval and safety controls.”

Note: E01 is marked in the source PDF as evidence to be replaced with actual interview evidence.